Unit - I

Two-Mark Questions

S. No.	Question	BT Level
1	Define a **data structure**. Give two examples.	BT1
2	What is an **Abstract Data Type (ADT)**? Give one example.	BT1
3	List any **two characteristics of an algorithm**.	BT1
4	Define **time complexity** and **space complexity**.	BT1
5	What is meant by **asymptotic analysis** of an algorithm?	BT1
6	Differentiate between **Big-O** and **Big-Ω** notations.	BT2
7	What is the time complexity of accessing an element in an array using its index?	BT1
8	Define a **one-dimensional array** with an example.	BT1
9	What is a **multi-dimensional array**? Give one example.	BT1
10	Write the general formula for calculating the **address of an element in a one-dimensional array**.	BT1
11	What is a **linked list**? State one advantage over an array.	BT1
12	Define a **singly linked list** and mention its basic components.	BT1
13	What is a **doubly linked list**? How does it differ from a singly linked list?	BT2
14	Define a **circular linked list**. How does it differ from a linear linked list?	BT2
15	List any **two basic operations** performed on a linked list.	BT1
16	What is the purpose of the **head pointer** in a linked list?	BT1
17	State the time complexity of **searching for an element** in a singly linked list in the worst case.	BT1
18	Mention any **two applications of linked lists**.	BT1
19	What is the advantage of a **doubly linked list** over a singly linked list?	BT2
20	Differentiate between a **linear linked list and a circular linked list**.	BT2

Sixteen-Mark Questions

S. No.	Question	BT Level
1	Explain the fundamental concepts, types, classification, and applications of data structures with suitable examples.	BT2
2	Explain Abstract Data Types (ADT), their characteristics, advantages, and implementation with suitable examples.	BT2
3	Describe the characteristics of an algorithm with suitable examples.	BT2
4	Explain time and space complexity analysis of algorithms. Discuss best-case, average-case, and worst-case complexities with suitable examples.	BT2
5	Explain asymptotic notations in detail. Discuss Big-O, Big-Ω, and Big-Θ	BT2

	notations and illustrate each with suitable examples and growth-rate comparisons.	
6	Analyse the time complexity of linear search, binary search, and simple sorting operations using asymptotic notation.	BT4
7	Explain one-dimensional arrays and their declaration, initialization, representation, and basic operations with suitable algorithms.	BT3
8	Explain multi-dimensional arrays with two-dimensional representation, row-major and column-major order, traversal, and applications.	BT3
9	Derive address calculation formulas for one-dimensional and two-dimensional arrays using row-major and column-major order with suitable examples.	BT4
10	Compare arrays and linked lists based on memory representation, access, insertion, deletion, searching, advantages, limitations, and applications.	BT4
11	Explain singly linked lists, including node representation and algorithms for creation, traversal, insertion, deletion, searching, and updating operations.	BT3
12	Develop algorithms for insertion and deletion in a singly linked list at the beginning, end, and specified position.	BT3
13	Explain doubly linked lists with node representation and algorithms for traversal, insertion, deletion, and searching.	BT3
14	Explain circular linked lists with their representation, traversal, insertion, deletion, and applications.	BT3
15	Compare singly, doubly, and circular linked lists based on structure, operations, advantages, limitations, and applications.	BT4
16	Design an appropriate linked-list-based solution for maintaining a dynamic student record system. Explain the data representation and algorithms for insertion, deletion, searching, updating, and traversal.	BT5
17	Explain the basic operations performed on linked lists. Develop algorithms and illustrate the steps involved in traversal, insertion, deletion, and searching using suitable examples.	BT3
18	Compare arrays and linked lists based on time, space, and basic operations with suitable applications.	BT4
19	Design a circular linked list for implementing a round-robin scheduling system. Explain node representation, traversal, insertion, deletion, and the working of the scheduling process with an example.	BT5
20	Evaluate the suitability of arrays, singly, doubly, and circular linked lists for frequently changing records based on memory, access, insertion, and deletion.	BT5

Unit II

Two-Mark Questions

S. No.	Question	BT Level
1	Define a **stack**.	BT1
2	What is the **LIFO principle**?	BT1
3	List any two basic operations of a stack.	BT1
4	What are **stack overflow** and **stack underflow** conditions?	BT1
5	How is a stack represented using an **array**?	BT2
6	What is the advantage of representing a stack using a **linked list**?	BT2
7	Define the **PUSH** operation.	BT1
8	Define the **POP** operation.	BT1
9	What is **expression evaluation**?	BT1
10	Convert the infix expression **A + B * C** into postfix notation.	BT2
11	Define a **queue**.	BT1
12	What is the **FIFO principle**?	BT1
13	Name the two ends of a queue.	BT1
14	Define **enqueue** and **dequeue** operations.	BT1
15	What is a **circular queue**?	BT1
16	State one advantage of a circular queue over a linear queue.	BT2
17	What is a **priority queue**?	BT1
18	Define a **deque (double-ended queue)**.	BT1
19	List any two applications of a **stack**.	BT1
20	List any two applications of a **queue**.	BT1

20 Sixteen-Mark Questions

S. No.	16-Mark Question	BT Level
1	Explain stack representation using arrays and linked lists with suitable algorithms.	BT2
2	Explain the basic stack operations PUSH and POP with suitable algorithms and examples.	BT2
3	Compare array-based and linked-list-based stack implementations based on memory, operations, advantages, and limitations.	BT4
4	Explain stack applications in expression evaluation with a suitable example.	BT3
5	Explain the procedure for converting an infix expression into postfix using a stack with a suitable example.	BT3
6	Convert a given infix expression into postfix and explain the stack operations involved.	BT3
7	Explain the evaluation of postfix expressions using a stack with a suitable example.	BT3
8	Explain queue representation and basic operations with suitable	BT2

	algorithms.	
9	Explain the implementation of a circular queue and its enqueue and dequeue operations with algorithms.	BT3
10	Compare linear queues and circular queues based on representation, operations, memory utilization, and applications.	BT4
11	Explain priority queues, their types, operations, and applications with suitable examples.	BT2
12	Explain deque and its types. Describe insertion and deletion operations at both ends with algorithms.	BT3
13	**Compare queue, circular queue, priority queue, and deque based on operations and applications.**	BT4
14	**Develop algorithms for enqueue and dequeue operations in a circular queue and illustrate them with an example.**	BT3
15	**Explain the applications of stacks and queues in real-world computing systems with suitable examples.**	BT3
16	**Evaluate the suitability of stacks, queues, and priority queues for different application scenarios.**	BT5
17	**Explain recursion and its relationship with stacks, using a suitable recursive algorithm as an example.**	BT2
18	**Develop a recursive algorithm for solving a suitable problem and explain its execution using a stack.**	BT3
19	**Compare iterative and recursive approaches based on execution, memory usage, advantages, and limitations.**	BT4
20	**Design a suitable stack or queue-based solution for a real-world application and explain its operations and implementation.**	BT5